

IP Addressing & Subnetting Technical Report

Comprehensive Solutions for Network Classification, Subnet Calculations, Enterprise Architecture, and Troubleshooting

Task 1: IP Address Classes and Real-World Examples

IPv4 addresses are historically categorized into five primary classes (A through E) based on their leading bits, which dictate the scale of the network versus the host allocation structure.

CLASS	LEADING BITS	FIRST OCTET RANGE	DEFAULT SUBNET MASK	REAL-WORLD EXAMPLE / USE CASE
Class A	0	1 – 126	255.0.0.0 (/8)	Large Telecom Core Router: Global enterprise or Tier-1 ISP core routing infrastructure managing millions of hosts.
Class B	10	128 – 191	255.255.0.0 (/16)	University Campus Network: Large institutional networks supporting tens of thousands of student and staff devices.
Class C	110	192 – 223	255.255.255.0 (/24)	Small Office / Home Office (SOHO) Router: Residential Wi-Fi routers connecting local smart home devices and laptops.
Class D	1110	224 – 239	N/A (Multicast)	IPTV Stream Server: Video-on-demand or live multicast distribution servers broadcasting to multiple client receivers simultaneously.
Class E	1111	240 – 255	N/A (Experimental)	Network R&D Testbed: Experimental protocols and research nodes reserved by IETF for future internet standards.

Task 2: Subnet Analysis for IP 192.168.10.45 / 255.255.255.0

Given the IPv4 address 192.168.10.45 with a subnet mask of 255.255.255.0 (equivalent to prefix length /24), we calculate the key network boundaries step-by-step.

Step 1: Calculate Network Address

Perform a bitwise logical AND operation between the IP address and the subnet mask.

```
IP: 11000000.10101000.00001010.00101101 (192.168.10.45)
Mask: 11111111.11111111.11111111.00000000 (255.255.255.0)
-----
Network: 11000000.10101000.00001010.00000000 (192.168.10.0)
```

Step 2: Calculate Broadcast Address

Set all host bits (the last 8 bits where the mask is 0) to binary **1**.

```
Network: 11000000.10101000.00001010.00000000 (192.168.10.0)
Inverted: 00000000.00000000.00000000.11111111 (Host bits wildcard)
-----
Broadcast: 11000000.10101000.00001010.11111111 (192.168.10.255)
```

Step 3: Determine Valid Host Address Range

Valid host addresses span from one number above the network address up to one number below the broadcast address.

First Host: Network address + 1 → 192.168.10.1

Last Host: Broadcast address - 1 → 192.168.10.254

Total Usable Hosts: $2^8 - 2 = 254$ hosts.

Task 3: Subnetting Plan for a Food Delivery Startup (10.0.0.0/24)

A startup like Zomato requires 4 separate isolated subnets for Delivery Partners, Restaurants, Customers, and Admins. Dividing the base 10.0.0.0/24 network into 4 equal subnets requires borrowing 2 additional host bits for subnetting ($2^2 = 4$ subnets).

Subnetting Parameters:

- Base Network: 10.0.0.0/24 (Subnet Mask: 255.255.255.0)
- Bits Borrowed: 2 bits → New Subnet Mask: /26 (255.255.255.192)
- Block Size (Hosts per Subnet): $2^{(32 - 26)} = 2^6 = 64$ addresses total (62 usable hosts per subnet).

USER GROUP	NETWORK ADDRESS	SUBNET MASK	USABLE HOST RANGE	BROADCAST ADDRESS
	10.0.0.0			10.0.0.63

USER GROUP	NETWORK ADDRESS	SUBNET MASK	USABLE HOST RANGE	BROADCAST ADDRESS
Delivery Partners		255.255.255.192 (/26)	10.0.0.1 - 10.0.0.62	
Restaurants	10.0.0.64	255.255.255.192 (/26)	10.0.0.65 - 10.0.0.126	10.0.0.127
Customers	10.0.0.128	255.255.255.192 (/26)	10.0.0.129 - 10.0.0.190	10.0.0.191
Admins	10.0.0.192	255.255.255.192 (/26)	10.0.0.193 - 10.0.0.254	10.0.0.255

Task 4: College WiFi Network Subnetting (172.16.0.0/16)

The network administrator needs to create at least 20 subnets from the base network **172.16.0.0/16** to service different campus buildings.

Step 1: Calculate Bits Needed for Subnets

We need to find the smallest integer n such that $2^n \geq 20$.

Since $2^4 = 16$ (insufficient) and $2^5 = 32$ (sufficient), we must borrow $n = 5$ bits from the host portion.

Step 2: Determine New Subnet Mask and Prefix

- Original Prefix: **/16**
- New Prefix: **16 + 5 = /21**
- New Subnet Mask calculation: **11111111.11111111.11110000.00000000** → **255.255.248.0**
- Total subnets created: **32** subnets (exceeds the requirement of 20).

Step 3: Calculate Available Hosts Per Subnet

With a **/21** prefix, there are **32 - 21 = 11** host bits remaining.

Total IP addresses per subnet = $2^{11} = 2,048$

Usable host addresses = $2^{11} - 2 = 2,046$ hosts per building subnet.

Task 5: Troubleshooting IP Address Error (**192.168.1.300**)

Error Identification:

The IP address **192.168.1.300** is invalid because IPv4 addresses consist of four octets, where each octet must be an integer strictly between **0** and **255** (representing 8 bits or values from **0** to **255**). The final octet **300** exceeds the maximum allowable value of **255**.

Corrective Solution & Valid Address Structure:

To connect successfully to the local network (assuming standard SOHO subnet mask **255.255.255.0**), the device must be assigned an IP address within the range **192.168.1.1** to **192.168.1.254**.

Example of a Valid IP Address: **192.168.1.30** (keeping the network prefix **192.168.1.** and choosing a valid host digit between 1 and 254).